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MJP ASSIGNMENT 1

Aim:- Write a java program to showcase classes, object creation, methods, constructors using student information.

Objectives:-

1. To study classes, objects and creation of objects of a class.
2. To study methods, return a value from the method and pass parameter to the methods
3. To study constructors and use of constructors
4. To study different access specifiers in Java & their use for encapsulation.

Theory:-

1. a) Class - It is a blueprint or template used to create objects. It defines properties and methods that an object will have.

b) Object - An object is an instance of a class. It represents a real world entity and occupies memory.

Example-

```
class Student {
    string name;
    int rollno;
}
Student s1 = new Student(); // object creation
s1.name = "Shreya";
s1.rollno = 59;
```

Student → Class S1 → Object of class Student

2. Syntax of a Method

```
accessSpecifier returnType methodName (parameters) {
    //method body
}
```

Ex1: Method with parameters and return value

```
class Calculator {
    int add (int a, int b) {
        return a+b;
    }
}
Calculator c = new Calculator();
int result = c.add (10, 20);
```

Ex 2: Method with No Return Value

```
class Message {
    void display () {
        System.out.println ("Welcome to Java");
    }
}
```

3. Use of Constructor

A constructor is a special method used to initialize objects.

Use

→ Initializes data members
→ Avoids writing separate initialization methods
→ Improves code readability

Example

```
class Student {
    string name; int rollno;
    Student (String n, int r) {
        name = n;
        rollno = r;
    }
```

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```
    void display() {
        System.out.println(name + " " + rollno);
    }
}
Student s1 = new Student ("Rahul", 102);
s1.display();
```

Platform:- Open Source Java Programming tools like Eclipse.

Conclusion:- Thus, studied what are classes, object creation, methods, constructors, access specifiers

FAQs:

Q1.) What is a class and object?

→ Class- A class is a blueprint or template that defines properties (variables) and behaviors (methods).
Object- An object is an instance of a class. It represents a real world entity and occupies memory.

Q2.) What is a constructor in Java?

→ A constructor is a special method used to initialize objects.

Features

→ Name is same as class name.

→ Has no return type.

→ Automatically called when an object is created.

Ex:
```
class Student {
    Student() {
        System.out.println("Object created");
    }
}
```

Q3) What is the default access specifier and what is its scope?

→ The default access specifier is used when no access specifier is specified.

Scope:

- Accessible within the same package
- Not accessible outside the package.

Example:

```
class Test {
    int data = 10;   // default access
}
```

Q4.) What are the different access specifier?

→ Java provides four access specifiers:

a) public

→ It is accessible everywhere

b) protected

→ It is accessible within package and subclasses

c) default

→ It is accessible within same package only.

d) private

→ It is accessible within same class only.

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